

Phase 8

Project Demonstration Phase

Demo walkthrough, sample outputs and links

1. Demonstration Overview

This phase presents the working AI StudyBuddy backend: a secure REST API that lets a student sign in, upload study material and instantly receive an AI-generated summary, flashcards, a quiz and a personalized study plan. The demo is run locally and driven from Postman.

2. Demo Materials

Item	Detail
Backend Development Demo	AI_Studybuddy_Backend_Demo.mkv
API Testing / Execution Demo	AI_Studybuddy_Execution_Demo.mkv
Demo video link	https://drive.google.com/file/d/11ShkBRp-qSXogvOqFgn4lnZa3CQok1GFr/view?usp=drive_link
Source code	Shared via the Code drive link in the project document

The demo link above was read from the project document (it was embedded as a scanned image), so please click it once to confirm it opens correctly.

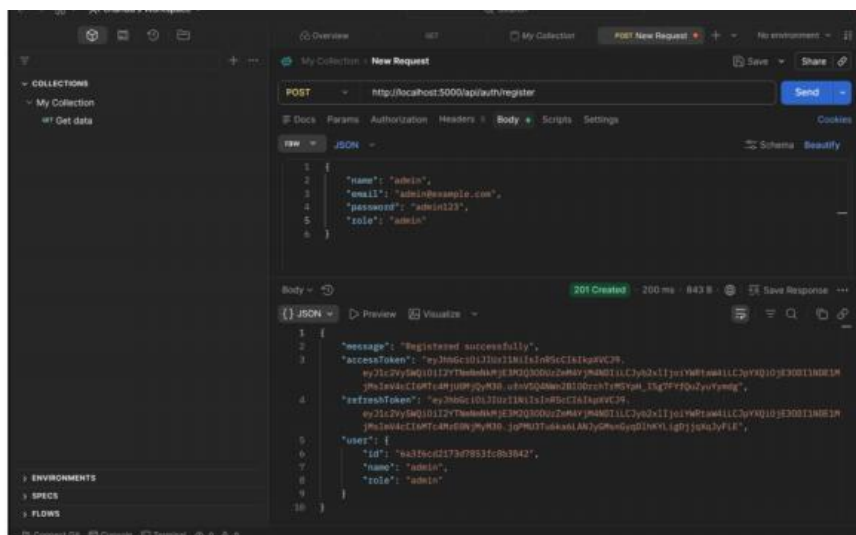
3. Pre-Demo Checklist

- MongoDB is running and .env has PORT, MONGO_URI, JWT secret and GEMINI_API_KEY.
- Dependencies installed (npm install) and server started (npm start).
- Postman open with the requests ready and a sample file such as 'Node.js Introduction.txt'.
- Console shows 'MongoDB connected' and 'Server running on port 5000'.

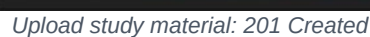
4. Demo Walkthrough (Suggested Script)

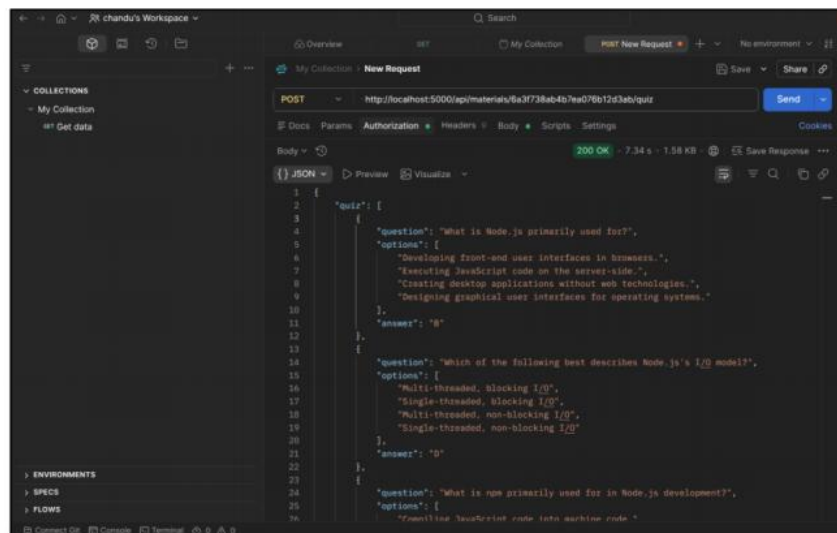
Step	What to show	What to say
1	Project structure in VS Code and the running server	Modular MVC backend: controllers, middleware, models, routes, utils. Server connects to MongoDB before listening.
2	POST /api/auth/register	Creates a user; password is bcrypt-hashed; JWT tokens returned (201 Created).
3	POST /api/auth/login	Authenticates and returns an access token used for all protected APIs (200 OK).
4	POST /api/materials/upload	Authenticated upload of a .txt file with a title; content stored in MongoDB (201 Created).
5	POST /api/materials/:id/summarize	Sends the material to Gemini and returns a concise summary.
6	POST /api/materials/:id/flashcards	AI creates question-answer flashcards for active recall.
7	POST /api/materials/:id/quiz	AI creates multiple-choice questions with options and the correct answer.
8	POST /api/materials/:id/study-plan	Given goal, hours per day and days, AI returns a day-by-day study plan.

5. Sample Outputs

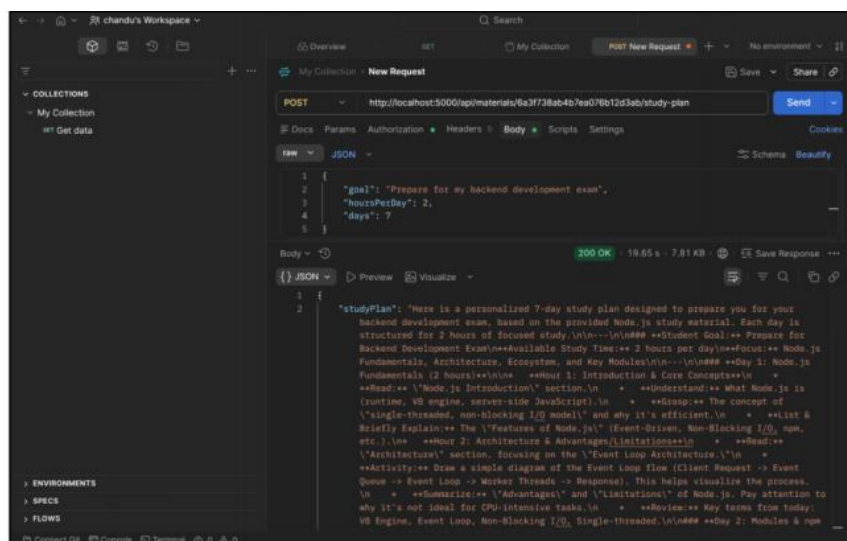


Register: 201 Created





AI quiz: 200 OK



AI study plan: 200 OK (7-day plan, 2 hours per day)

6. Key Talking Points

- **Problem solved:** students like Rahul spend hours summarizing notes and making practice questions; AI StudyBuddy automates this.
- **Security:** JWT authentication, hashed passwords, role-based access for students and admins.
- **AI integration:** content is generated dynamically by Google Gemini from the student's own material, not from fixed templates.
- **Architecture:** modular MVC design that makes it easy to add more AI features.
- **Response times:** login and upload take ~0.1 to 0.2 s; AI generation takes ~5 to 20 s depending on the task.

7. Expected Outcome

By combining artificial intelligence with a secure backend architecture, AI StudyBuddy improves the learning experience: manual preparation time is reduced through automated content generation so students can focus on understanding concepts, and the modular design supports future AI-powered capabilities with minimal changes.